Naksha (নকশা)

Inter-Agency Land Data Interoperability & Canonical Spatial Engine

One parcel. One identity. Every land office looking at the same truth.

1 The problem: records exist, but they do not agree

Bangladesh does not suffer from a lack of land records. It suffers from **too many records that never meet**. A single plot is recorded by at least six state bodies, each in its own system and under its own key, and none of them can see the others at the moment an officer acts.

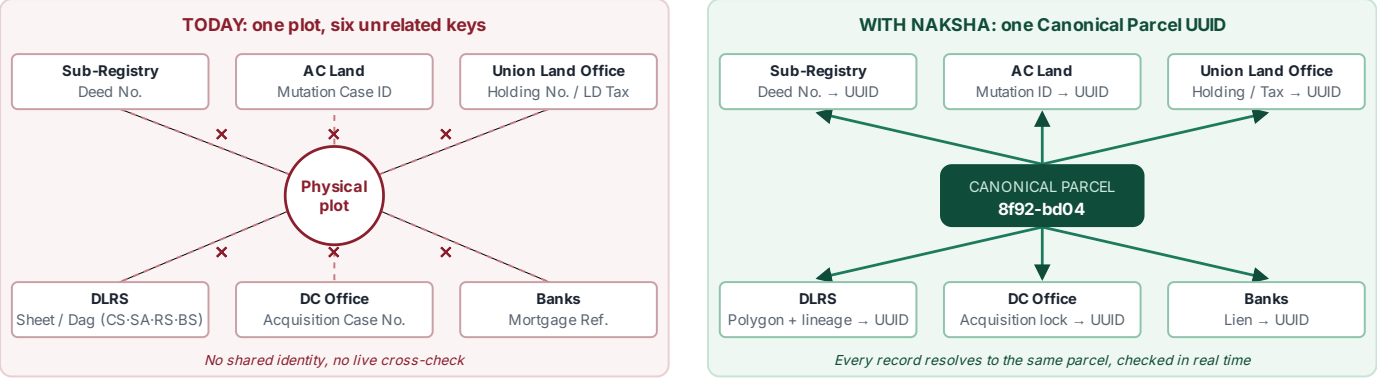


Figure 1. Today each agency identifies a plot by a different key. Naksha resolves all of them to one canonical identity.

The consequences are well known, and they are all the same failure seen from different offices:

- A deed is registered but no mutation is ever opened, so the same plot can be **sold a second time**.
- LD Tax is accepted from someone who is not the recorded owner, and the receipt later becomes evidence of possession.
- Acquisition compensation is paid to an **outdated record holder** because the mutation trail was invisible to the DC office.
- The same parcel is **mortgaged at several banks**, or sold while encumbered, because liens are not shared.
- Paper decimals exceed the surveyed polygon, or quietly overlap Khas land, because nobody compares the deed against DLRS geometry.

2 The solution: a federated overlay, not another portal

> **Naksha** is a Government-to-Government interoperability engine. It does not replace any ministry's software. It **connects** the systems that already exist through a canonical parcel identity, and shows each officer what the other agencies know, inside the tool they already use.

Canonical Parcel Model

One persistent parcel UUID links deeds, mutations, LD Tax holdings, DLRS polygons, acquisition locks and bank liens, across all four survey eras (CS, SA, RS, BS).

Zero-integration overlay

A Chrome extension adds a secure side panel on top of the portals officers already use. No legacy code is modified, so rollout does not wait for a system overhaul.

Deterministic integrity engine

Five fixed rules and PostGIS geometry checks run whenever a plot is touched. Every result is explainable and sealed in a tamper-evident ledger.

Design principles

- **Non-invasive.** Federated adapters read from agency systems; the source of record stays with each ministry.
- **Explainable.** Legal decisions are made by rules and geometry, never by opaque models.
- **Role-scoped.** An officer sees only what their ministry and jurisdiction are entitled to see.
- **Evidence-grade.** Every query and decision is hash-chained so it can stand up in an audit or a court.

Naksha at a glance

6	4	5	4	0
agency systems linked through one parcel UUID	survey eras reconciled: CS, SA, RS, BS	deterministic integrity rules on every touch	map layers: history, claim, ground truth, conflict	changes to legacy ministry software

3 The officer experience: the Naksha Chrome extension

A government employee opens the portal they use every day: deed registration, e-Mutation, LD Tax or acquisition. The Naksha extension recognises the page, reads the plot identifiers on screen, asks the officer to sign in with their official credentials, and opens a **side panel** with the full picture of that plot: land history, GIS map, tax map, liens and a clearance verdict.



Figure 2. From the officer opening a portal to a verified, role-scoped view of the plot.

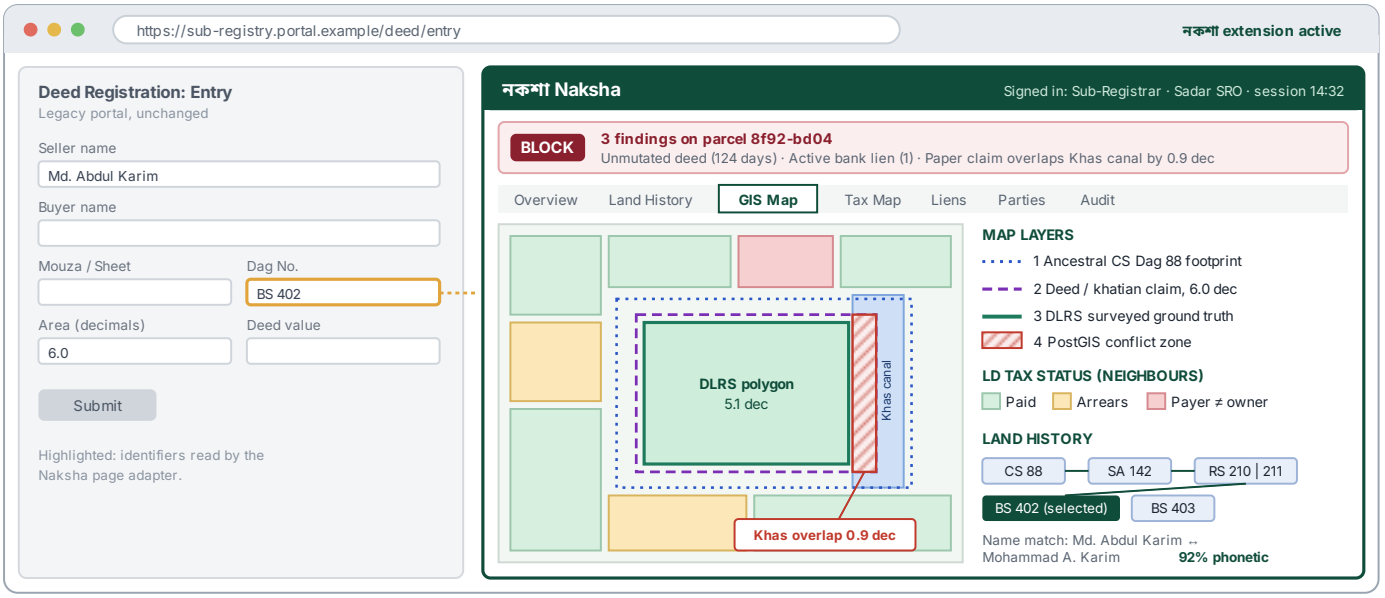


Figure 3. Illustrative side panel on a Sub-Registrar's deed-entry screen (synthetic data). The GIS Map tab overlays four survey layers and LD Tax status.

What the side panel shows

- **Verdict.** CLEAR, REVIEW or BLOCK, with each finding linked to the records behind it.
- **Land history.** The unbroken CS → SA → RS → BS lineage, with fractional shares carried across generations.
- **GIS map.** Ancestral footprint, paper claim, DLRS ground truth and the computed conflict zone.
- **Tax map.** LD Tax status of the parcel and its neighbours, who paid, and whether the payer matches the khatian owner.
- **Liens and locks.** Active bank mortgages and DC acquisition notices on the same parcel UUID.
- **Audit.** A signed trail of who looked at this plot and what they did.

One plot, five different views

Officer	Host portal	What Naksha surfaces and enforces
Sub-Registrar	Deed registration	Pre-registration clearance: seller vs latest mutation, unmutated earlier deed, bank lien, paper area vs DLRS polygon.
AC Land officer	e-Mutation	Verified deed copy, full survey lineage, name-match score and boundary check before a mutation is approved.
Union Land officer	LD Tax portal	Payer integrity alert, Nal/Viti class conflict, and the tax map for the surrounding plots.
DC acquisition officer	Acquisition portal	Project corridor overlaid on DLRS polygons, with the current legal owner shown so compensation reaches the right person.
Banks, courts, citizens	Verification API / web	Lien lock across banks, hash-chained audit evidence for courts, and a "check before you buy" lookup.

4 Backend architecture

Naksha is a layered, event-driven backend. Agency data is normalised at the edge, resolved to a canonical identity, checked geometrically and legally, and sealed in a ledger before anything reaches the officer's panel.

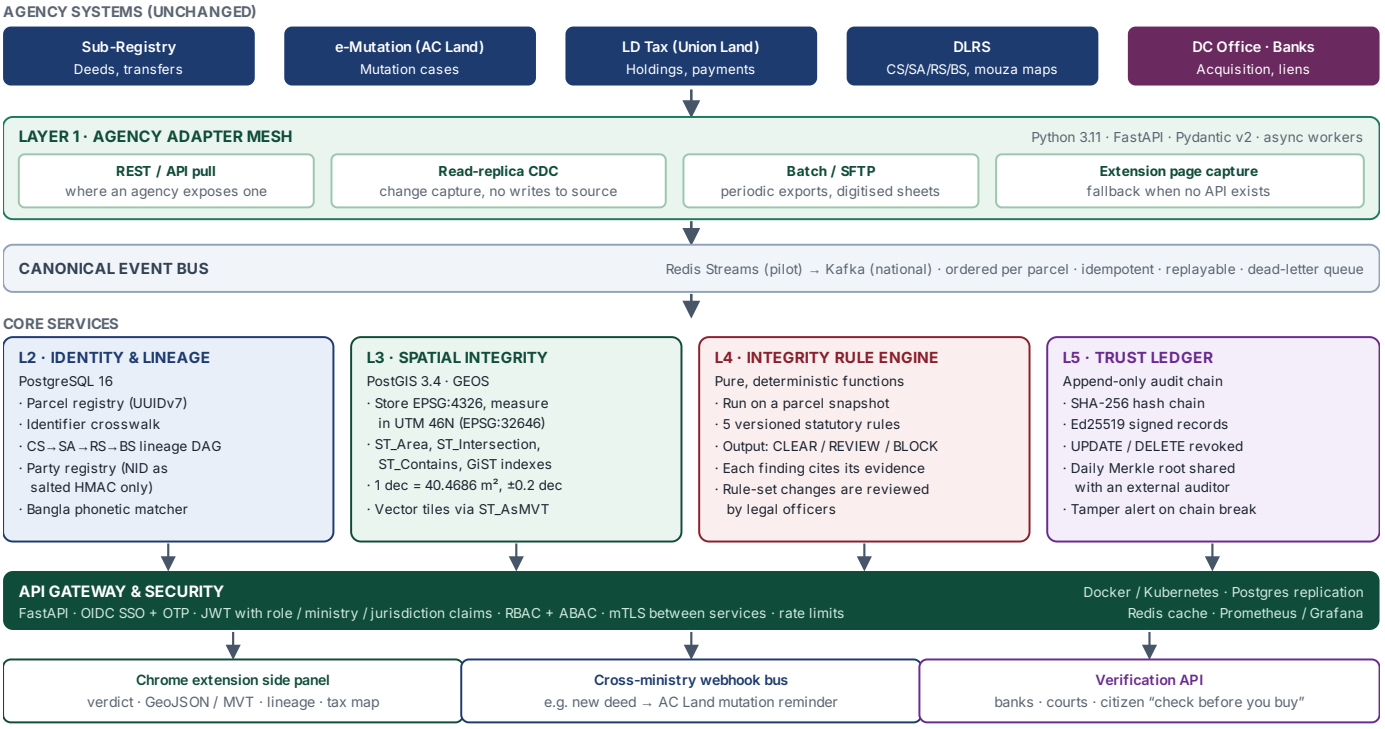


Figure 4. Naksha backend: five layers behind a secured gateway, serving the extension, ministry webhooks and external verifiers.

Key design decisions

Concern	How Naksha handles it
Heterogeneous sources	Each agency gets a plug-in adapter that maps its schema into canonical events. Adding an agency means writing one connector, not changing the core. Ingestion is idempotent and replayable, so a failed sync never corrupts state.
Parcel identity	A crosswalk table maps (agency, key type, key value) to a parcel UUID with validity dates and match confidence. Survey subdivisions are stored as a DAG, and the **sum of child shares must equal the parent share** (within tolerance) or the lineage is flagged.
Bangla name matching	Unicode normalisation, honorific stripping (মোঃ, মোছাঃ), phonetic keys and edit distance produce a visible score such as "92% · Phonetic equivalent". Scores below the threshold go to a human reviewer. No black-box model decides ownership.
Geometry	Polygons are validated (ST_MakeValid), stored in WGS84 and measured in a metric projection. Paper decimals are compared with polygon area, and intersections with Khas land, corridors and neighbouring plots are returned as a conflict polygon the map can draw.
Privacy and access	Need-to-know by role and jurisdiction. NIDs are stored only as salted HMACs, sensitive fields are encrypted at rest, and the extension receives only the fields its role is entitled to.

Request lifecycle (design target: verdict in under one second at p95)

1 Token and role validated at the gateway → **2** identifiers resolved to the parcel UUID → **3** snapshot assembled (lineage, parties, tax, liens, geometry) → **4** spatial checks run → **5** rules evaluated, verdict produced → **6** record sealed in the ledger and the response returned as JSON plus map tiles.

5 Validation on a synthetic federation

Synthetic federation. Naksha uses no live government databases and no real citizen names, NIDs, phone numbers or land records. Four isolated mock schemas (Sub-Registry, AC Land, LD Tax, DLRS) are filled by a deterministic seeder, `seed_synthetic.py`, which reproduces four scenarios end to end:

- CLEAR Clean parcel: unbroken CS→BS chain, synchronised tax and mutation, 6.0 dec polygon.
- BLOCK Unmutated double-sale hazard: deed registered four months earlier, no mutation.
- REVIEW Adverse possession attempt: LD Tax paid by a non-owner with a Nal/Viti conflict.
- BLOCK Spatial encroachment: 6.0 dec claim vs 5.1 dec polygon crossing a Khas canal strip.

6 Fraud prevention is a by-product of integrity, not a bolt-on

Naksha is not built as a fraud detector. Fraud thrives in the gaps between agencies, so closing those gaps removes most of the opportunity. The five rules below simply make the gaps visible to the officer at the exact moment they could be exploited.

Rule	Fires when	Gap it closes
`RULE_UNMUTATED_DEED`	A deed was registered more than 30 days ago with no mutation opened at AC Land.	Double-selling before a second deed is registered.
`RULE_TAX_OWNER_MISMATCH`	LD Tax is offered by someone who is not in the current mutated khatian or a verified inheritance chain.	Receipts used to manufacture possession claims.
`RULE_CLASS_CONFLICT`	Khatian class (e.g. agricultural Nal) differs from the class assessed in LD Tax (e.g. Viti).	Silent change of land use and revenue evasion.
`RULE_ACQUISITION_OR_MORTGAGE_LOCK`	An acquisition notice or bank lien is active on the parcel UUID.	Transfer of encumbered land, double mortgage.
`RULE_GEOMETRIC_OVERLAP`	PostGIS finds intersection with Khas land, forest or wetland, or a neighbouring plot; or paper area exceeds the polygon beyond ±0.2 dec.	Encroachment on public land and inflated deed areas.

7 How Naksha is different

Common approach	Where it falls short	What Naksha does instead
Separate digital portals per office	Each shows only its own records; the same plot has a different key in every system.	Canonical parcel UUID and live cross-agency checks at the point of action.
New national registry or full platform replacement	Multi-year migration, new training, and legacy data problems remain.	Overlay and adapters: officers keep their portals, agencies keep ownership of data.
AI fraud scoring	Opaque probabilities that a legal officer cannot cite or defend.	Deterministic rules and geometry; every flag points to records, and every name match shows its score.
Stand-alone GIS viewers	Show a map, but do not reconcile paper claims, history and ground truth.	Four-layer reconciliation with a computed conflict polygon and tax overlay.
Audit logs in the same database	An administrator can edit history without trace.	Hash-chained, signed, externally anchored ledger that breaks visibly if altered.

8 Rollout path

Phase 1 · Pilot
One upazila: Sub-Registry, AC Land and Union Land offices on the extension, with read-only adapters.

Phase 2 · District
Add DLRS polygons, DC acquisition and bank lien feeds; turn on cross-ministry webhooks.

Phase 3 · National
Scale across 64 districts on Kafka and replicated PostGIS; open the verification API to banks and courts.

9 Team and mentors

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